

ECCB: Department of Ecology and Conservation Biology

ECCB 689 Modeling Ecological Niches and Species' Distributions

Lab 04: Environmental Data Audit and Processing

Today you will inspect environmental raster layers, restrict them to a study area, extract values at retained occurrences, and examine how transformation and spatial grain change a predictor. You will then inspect the environmental files that you downloaded for your project.

Learning goals

- Read raster layer names, coordinate reference system, spatial extent, grain, and summary.
- Crop and mask environmental layers to a geographic hypothesis about M.
- Extract environmental values at occurrence locations and save a table.
- Explain how a log transformation and a coarser grid alter the values available for analysis.
- Audit downloaded project layers and use the results in the Friday Data Plan.

What you need

- Your course R Project and private GitHub repository.
- A completed or working Lab 3 script in analysis/03_environmental_space_background.R.
- R packages sf and terra installed.
- At least three downloaded project environmental layers saved in data/. GeoTIFF (.tif or .tiff) and NetCDF (.nc) files work with terra.

Part A – Guided workflow

Create your own Lab 04 script

1. **Create a new R script** in your analysis/ folder and save it as analysis/04_environmental_data_audit.R.
2. **Type the code blocks below into your own script** as you work through it.
3. **Run from the top** each time you make a substantial change to check for any coding errors or issues with repeatability.

Load the Lab 03 objects

Source Lab 3 objects so you can recreate your retained occurrences, raster stack, and broad-M geometry. This helps with workflow and prevents you from needing to copy objects manually from one script into another.

```
library(sf)
library(terra)

source("analysis/03_environmental_space_background.R")
```

Inspect the environmental raster

The supplied stack contains annual mean temperature (bio1_c, degrees C) and annual precipitation (bio12_mm, mm) from WorldClim 2.1 for 1970-2000. The 10 arc-minute grid is deliberately coarse so the classroom workflow runs quickly.

```
env
names(env)
crs(env)
res(env)
ext(env)
global(env, c("min", "max", "mean"), na.rm = TRUE)
```

Interpret the metadata: a raster is a grid of numbers associated with metadata including the definition of the variables, units, time period, coordinate system, spatial grain, and geographic extent all affect what it can represent.

Restrict layers to your broad M hypothesis

crop() limits processing to the polygon's rectangular extent. mask() sets cells outside the M polygon to NA. This changes the environmental domain you are describing.

```
env_broad_lab4 <- mask(
  crop(env, vect(m_broad)),
  vect(m_broad)
)

png(
  "output/04_biol_broad_M.png",
  width = 1800, height = 1400, res = 180
)
plot(env_broad_lab4[["bio1_c"]],
  main = "Annual mean temperature within broad M")
plot(vect(m_broad), add = TRUE, border = "firebrick3", lwd = 2)
points(vect(occ_sf), pch = 16, cex = 0.45)
dev.off()
```

Extract values at occurrence locations

Extraction pulls the cell value beneath each retained point and connects geographic observations to environmental space. The table you save is a useful record of exactly what environmental values were associated with each occurrence.

```
occ_env_lab4 <- extract(env_broad_lab4, vect(occ_sf))

occ_env_lab4 <- cbind(
  occ[, c("record_id", "species",
    "decimalLongitude", "decimalLatitude")],
  occ_env_lab4[, c("bio1_c", "bio12_mm")])

head(occ_env_lab4)
summary(occ_env_lab4[, c("bio1_c", "bio12_mm")])
colSums(is.na(occ_env_lab4[, c("bio1_c", "bio12_mm")]))
```

Check for missing values: an NA can mean a point falls outside the raster, the raster has no data at that location, or the coordinate systems do not align. Do not remove missing values until you know why they are NA.

Transform a predictor

A transformation is usually performed to make a heavily skewed distribution easier to inspect or model, but it also changes the scale and should be documented within your workflow. Precip is often strongly right-skewed, meaning that many locations receive relatively low to moderate precip, while a smaller number of locations receive very large amounts of precip. A log transform compresses the high end of the distribution and spreads out differences among lower values. This can make patterns easier to visualize and can make relationships with biological responses easier to model.

A log transformation can also make biological sense when relative differences matter more than absolute differences. For example, an increase from 10 mm to 20 mm represents a doubling of precipitation and may have a much larger biological effect than an increase from 2000 mm to 2010 mm, even though both changes are 10 mm. On a log scale, proportional changes are represented more similarly across the range of precipitation values. Use `hist()` to check the distribution of your dataset to determine if you think it requires a transformation. Retain the raw layer and use a transformation when you can explain why it is necessary.

```
precip_log10 <- log10(env_broad_lab4[["bio12_mm"]])
names(precip_log10) <- "bio12_log10"

occ_env_lab4$bio12_log10 <- log10(occ_env_lab4$bio12_mm)

hist(env_broad_lab4[["bio12_mm"]],
     main = "Annual precipitation", xlab = "mm")
hist(precip_log10,
     main = "Log10 annual precipitation", xlab = "log10(mm)")

write.csv(occ_env_lab4, "output/04_occurrence_environment_values.csv",
          row.names = FALSE)
```

Explore spatial grain

A coarse raster summarizes more area in each cell. The operation below creates cells twice as wide and twice as long, then compares the values extracted at the same occurrence locations.

```
env_coarse <- aggregate(env_broad_lab4, fact = 2, fun = mean)

res(env_broad_lab4)
res(env_coarse)

occ_env_coarse <- extract(env_coarse, vect(occ_sf))

resolution_comparison <- data.frame(
  record_id = occ$record_id,
  bio1_fine_c = occ_env_lab4$bio1_c,
  bio1_coarse_c = occ_env_coarse$bio1_c,
  bio12_fine_mm = occ_env_lab4$bio12_mm,
  bio12_coarse_mm = occ_env_coarse$bio12_mm
)

write.csv(resolution_comparison,
          "output/04_resolution_extraction_comparison.csv", row.names = FALSE)
```

Important distinction: aggregation changes the information content by averaging cells. Resampling can make grids line up, but it does not restore the local variation lost when a layer was aggregated.

Make grids align when needed

A raster stack requires compatible coordinate systems, extent, resolution, and cell alignment. This code gives the coarse layers the fine grid's geometry. It does not make the coarse layers more detailed.

```
env_coarse_resampled <- resample(  
  env_coarse,  
  env_broad_lab4,  
  method = "bilinear"  
)  
  
res(env_coarse_resampled)
```

Pause and interpret

1. **Compare the fine and coarse values** for several occurrence records. Do they match? Why?
2. **Explain the transformation** in one sentence: what changed when precipitation moved from mm to $\log_{10}(\text{mm})$?
3. **Identify one issue** that could make your project's occurrence records and environmental layers incompatible.

Part B Independent project layer audit

You were asked to download at least three environmental predictors for your project (see Week 6 slides). Use those files now. This short audit should help you write the Environmental Predictors and Data Compatibility sections of your Data Plan.

Load your project layers

Replace the file name with one of your own .tif, .tiff, or .nc files inside data/. Use an informative object name in your script. Add scripting to load all of your project predictors. Attempt to combine compatible layers into a multi-layer raster object. If layers are not compatible, diagnose the mismatch and document what processing would be needed before combining them.

```
project_layer1 <- rast("data/replace_with_your_layer.tif")  
# add code to load in project_layer2 and project_layer3  
  
names(project_layer1)  
crs(project_layer1)  
res(project_layer1)  
ext(project_layer1)  
  
# inspect the range and distribution of values  
global(project_layer1, c("min", "max", "mean"), na.rm = TRUE)
```

```
#plot histograms
hist(project_layer1)

png("output/04_project_layer1_preview.png",
     width = 1600, height = 1200, res = 180)
plot(project_layer1)
dev.off()

# repeat the metadata checks, summary, hist, and plot for each predictor

# combine multiple compatible layers, only works if layers are compatible
project_env <- c(project_layer1, project_layer2, project_layer3)
```

Complete these four tasks

1. **Save your own Lab 04 script** as analysis/04_environmental_data_audit.R. It should reproduce the guided course example and include the project-layer code with file paths.
2. **Inspect all downloaded environmental layers** and save plots into your output folder.
3. **Build the predictor table for your Data Plan** using approximately three to six candidate predictors. For each, document source and version, units, resolution, time period, direct/indirect/resource role, biological rationale, and one processing concern.
4. **Write a short README Lab 04 section** answering the prompts on the next page. Keep it concise and point to your output files or metadata as evidence.

If a project file will not load, note its format and the error in your README. Do not substitute an unrelated variable simply because it is easier to read.

Project predictor audit

Use this table as a planning aid for the environmental-predictor table in your Data Plan. Complete a row for each of your three to six candidate predictors. The source metadata comes from the provider; the ecological rationale comes from your question and organism.

Predictor	Source and version	Units	Grain and time	Role and rationale

com

References

- Fick, S. E., and Hijmans, R. J. (2017). WorldClim 2: New 1-km spatial resolution climate surfaces for global land areas. *International Journal of Climatology*, 37, 4302-4315.
- Peterson, A. T., Soberon, J., Pearson, R. G., Anderson, R. P., Martinez-Meyer, E., Nakamura, M., and Araujo, M. B. (2011). *Ecological Niches and Geographic Distributions*. Princeton University Press. Chapter 6.
- WorldClim 2.1 historical climate data: <https://www.worldclim.org/data/worldclim21.html>

Academic integrity: the submitted code, reasoning, and interpretation must be your own and understandable to you. Briefly acknowledge any AI use in a script comment or README note.

README Lab 04 prompts

1. **Describe the supplied course raster stack.** State the variable definitions, units, time period, coordinate system, extent, and spatial grain.
2. **Explain extraction.** What does one row of output/04_occurrence_environment_values.csv represent? What would an NA require you to investigate?
3. **Explain one data manipulation.** Choose either the log10 precipitation transformation or the fine versus coarse comparison. State what changed and why the change matters.
4. **Audit your project predictors.** Identify their source, units, resolution, time period, ecological role, and any possible mismatch or uncertainty. Refer to the predictor table in your Data Plan.

Assignment 4 submit Friday by midnight

Submit your private GitHub repository link in Canvas. Also submit the separate Data Plan PDF through the Data Plan assignment. Lab 04 is intentionally brief because the Data Plan is due the same day, but I still expect high attention to detail and increasing independence in understanding each piece of your code.

Reproducible Lab 04 script: analysis/04_environmental_data_audit.R runs from project root.

Course-example outputs: the broad-M temperature map, occurrence-environment table, and resolution extraction comparison.

Project environmental-layer previews: plots of all downloaded project environmental layers saved in the output folder.

README Lab 04 section: concise answers to the four prompts above.

Version history: commits show the Lab 04 work and the repository is pushed to GitHub.